

DYNAMIC ATTACKER
(Exploitations & Adaptation)

CRITICAL ASYMMETRY

STATIC DEFENDER
(Detection & Reactive Countermeasures)

Attack Adaptation Loop

RQ1: Exploitation & Generation (Phishing Lifecycle)

(A) NINE Stages Generation

S1 Basic Instruction

S2 Role-Framed Prompting

S3 Multiturn Task Decomposition

S4 Scenario driven Adaptation

S5 Personalization for Credibility

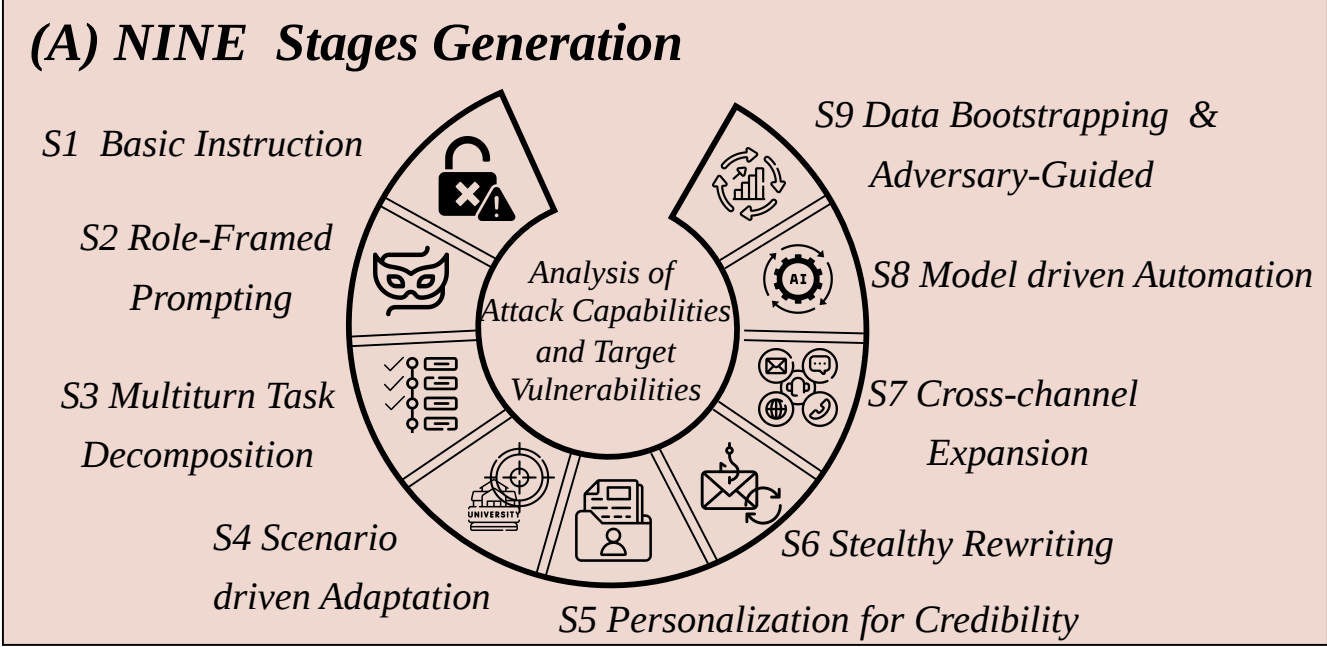
S6 Stealthy Rewriting

S7 Cross-channel Expansion

S8 Model driven Automation

S9 Data Bootstrapping & Adversary-Guided

Analysis of Attack Capabilities and Target Vulnerabilities



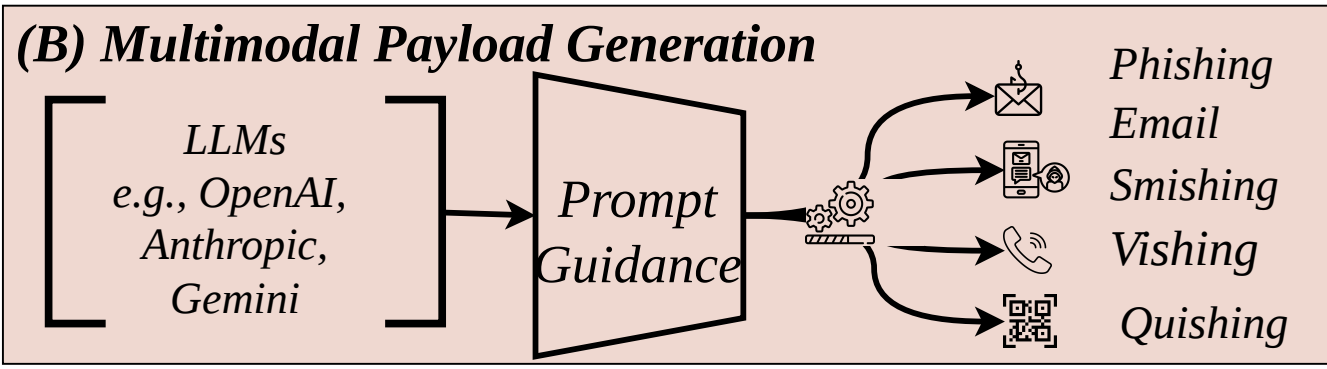
(B) Multimodal Payload Generation

The diagram illustrates the process of generating multimodal payloads. On the left, a box labeled "LLMs" contains the text "e.g., OpenAI, Anthropic, Gemini". An arrow points from this box to a trapezoidal box labeled "Prompt Guidance". From the right side of the "Prompt Guidance" box, an arrow points to a gear icon. From the gear icon, four arrows branch out to the right, each pointing to a different icon: an envelope (Phishing), a smartphone (Email), a smartphone with a lock icon (Smishing), a telephone receiver (Vishing), and a QR code (Quishing). The text labels for these attack types are listed to the right of their respective icons.

LLMs
e.g., OpenAI,
Anthropic,
Gemini

Prompt Guidance

Phishing
Email
Smishing
Vishing
Quishing



RQ2: Characterization & User Effects (Taxonomy)

*Payloads
Map to
Taxonomy*

(A) Textual Traits		{ Coherence Action-Centered
(B) Social Engineering		{ Authority Scarcity Urgency Consistency Reciprocity
(C) Model		[Scale

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*Guides
Defense
Design*

RQ3: Defense & Detection (Frameworks)

(A) Machine Detection

The diagram illustrates the machine detection process. A horizontal arrow represents the flow of information. On the left, the text "Before Generation Check" is positioned below the arrow. Above the arrow, there is a funnel icon with a checkmark inside it. The arrow points to a central box labeled "LLM". A vertical dashed line passes through the center of the "LLM" box. To the right of the "LLM" box, the text "After Generation Check" is positioned below the arrow. Above the arrow, there is a bell icon with a checkmark inside it. In the top right corner, there is a shield icon with a checkmark inside it. The text "Prompt-level Before Check" is located above the funnel icon, and "Output-level After Check" is located above the bell icon.

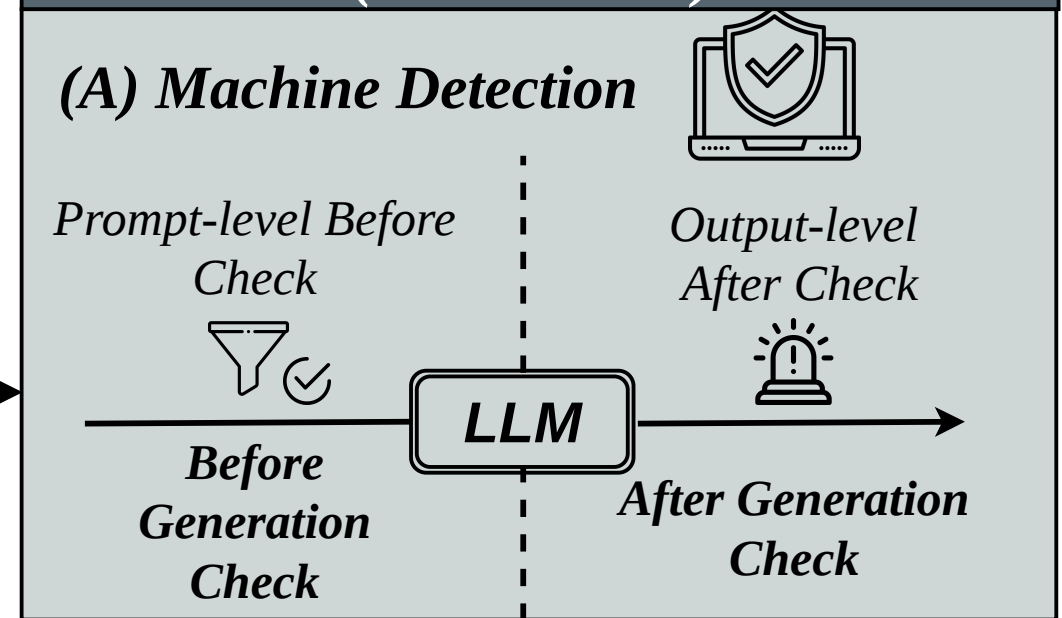
Prompt-level Before Check

Output-level After Check

Before Generation Check

LLM

After Generation Check



(B) Human-Centric Defense

The diagram illustrates a three-stage process for human-centric defense. It begins with 'Perception', represented by an icon of a human head profile with an eye. An arrow points to 'Cognition', represented by an icon of a brain and a gear. A second arrow points to 'Reaction', represented by an icon of a lightbulb with a checkmark inside. The labels 'Perception', 'Cognition', and 'Reaction' are written below their respective icons.

Perception *Cognition* *Reaction*

